Supplementary information for: TorchRef: An open-source PyTorch Framework for Crystallographic Refinement

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Reading guide

This supplement is in two parts. The first part presents the four **Supplementary figures** referenced from the main manuscript: the loss-weight calibration sweep (Fig. S1), the per-resolution TorchRef–reference R-factor comparison (Fig. S2), the cross-scorer validation (Fig. S3), and the single-core wall-clock runtime comparison (Fig. S4). The second part, **Supplementary methods**, is an algorithmic reference for readers who wish to understand or extend TorchRef's internals: parameterisations, structure factor calculation, X-ray target functions, restraints, the AMBER target, the multi-dataset infrastructure, loss aggregation, and the optimisation cycle.

Supplementary figures

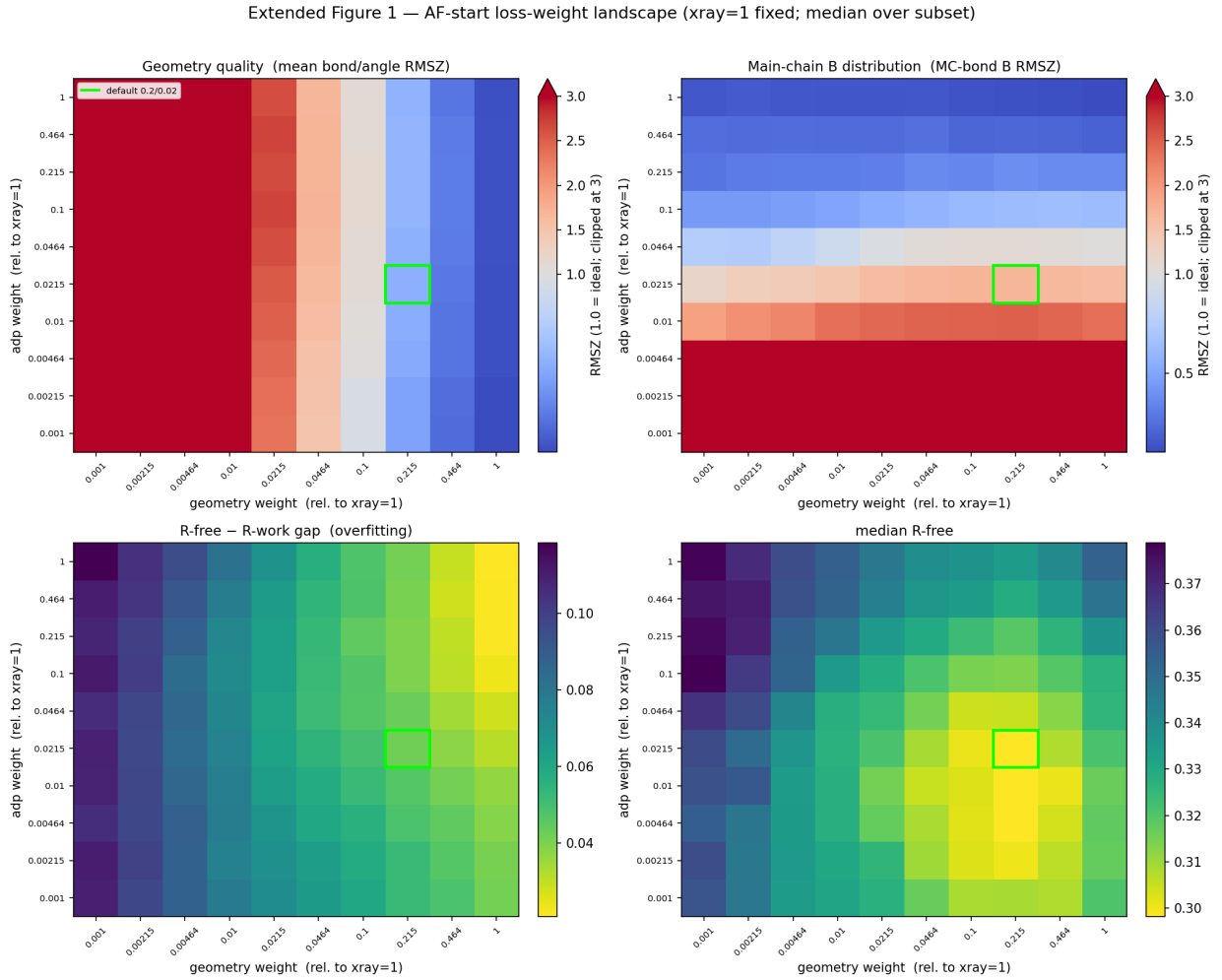


Figure S1: Loss-weight landscape on the AlphaFold benchmark, exploring the $(w_{\text{geometry}}, w_{\text{adp}})$ plane at fixed $w_{\text{xray}} = 1$. Each heat-map cell is the median refinement outcome for a stratified subset of structures refined at that weight setting; the green box marks the TorchRef default $(w_{\text{geometry}}, w_{\text{adp}}) = (0.2, 0.02)$. Top-left: median bond/angle RMSZ (geometry quality). Top-right: main-chain bonded-B RMSZ (ADP distribution). Bottom-left: median R_{work}–R_{free} gap (overfitting). Bottom-right: median R_{free}. The default sits in a flat region of the (R_{free}, R-gap, RMSZ) landscape; per-structure tuning within this neighbourhood is the expected workflow.

Extended Figure 2 — R-factor gap vs resolution (AF-start, PHENIX-scored)

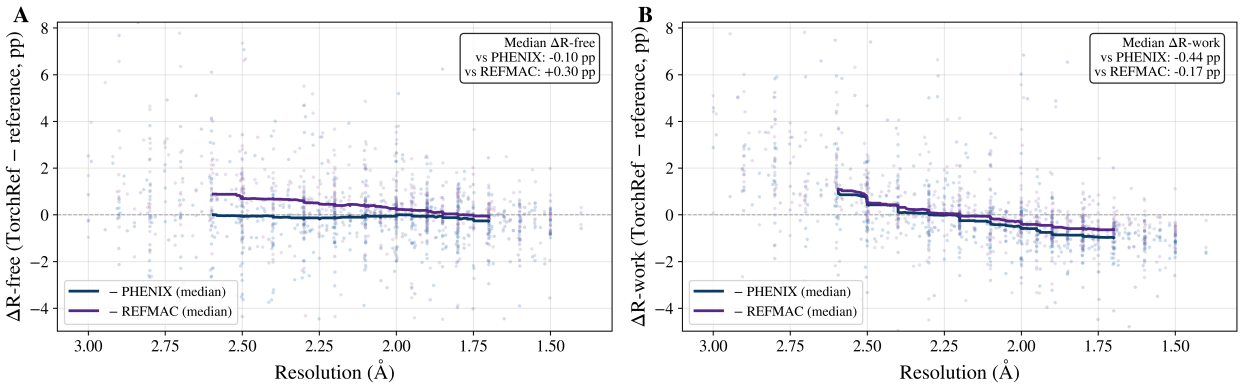


Figure S2: Per-structure TorchRef−reference R-factor differences as a function of resolution on the AlphaFold benchmark ($n = 725$ structures, $d_{\min}$ range 1.40–3.00 Å), with a light scatter and a bold running-median line for each reference program: Phenix (blue) and REFMAC5 (purple). A positive ΔR indicates that TorchRef's R-factor is higher than the reference. Panel A: ΔR_{free} vs resolution. Overall medians are −0.10 percentage points (pp) vs Phenix and +0.30 pp vs REFMAC5, consistent with the cross-engine R_{free} agreement reported in the main text. Panel B: ΔR_{work} vs resolution. Overall medians are −0.44 pp vs Phenix and −0.17 pp vs REFMAC5.

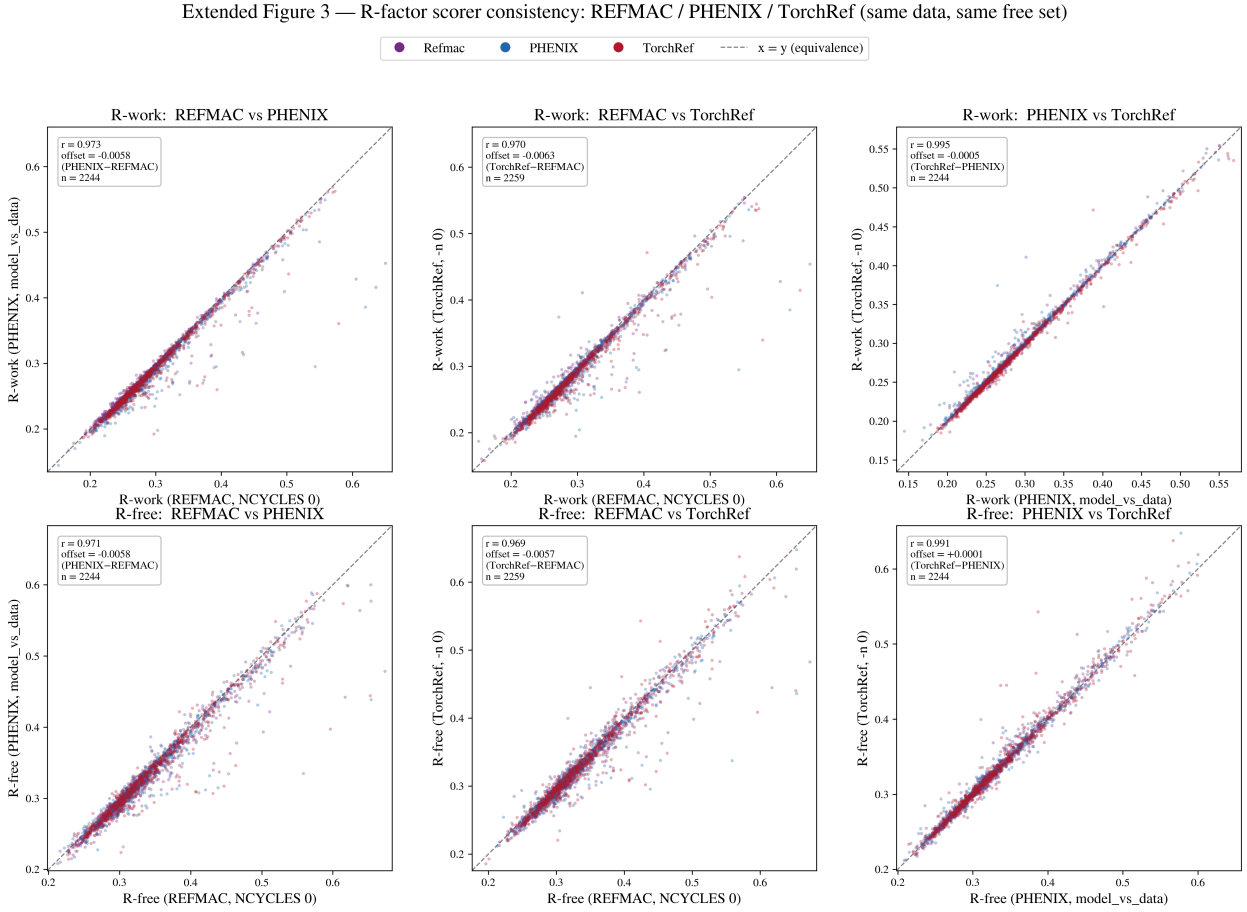


Figure S3: R-factor scorer cross-consistency on the AlphaFold benchmark. Every refined model (`refmac5`, `phenix.refine`, TorchRef; coloured by the engine that produced it) was independently re-scored against identical structure factors and free flags by three R-factor scorers operating in zero-cycle mode (scaling and bulk-solvent fit only; coordinates and ADPs untouched): REFMAC5 with `NCYCLES 0`, `phenix.model_vs_data`, and `torchref.refine -n 0`. Top row: R_{work}. Bottom row: R_{free}. Each panel shows pairwise scatter between two scorers, with Pearson correlation r, signed median offset (scorer$_{\text{y}}$ − scorer$_{\text{x}}$), and sample count annotated. The dashed line is the equivalence diagonal $x = y$. Phenix and TorchRef show very close agreement ($r = 0.995$ for R_{work}, $r = 0.991$ for R_{free}), with signed median offsets of -0.0005 and $+0.0001$, respectively. REFMAC5 vs TorchRef gives $r = 0.970$ for R_{work} and $r = 0.969$ for R_{free}, with median offsets of -0.0063 and -0.0057. REFMAC5 and Phenix give $r = 0.973$ for R_{work} and $r = 0.971$ for R_{free}, with offsets of -0.0058 and -0.0058. The three scorers measure equivalent R-factors to within a small systematic offset across all engine pairings, supporting the choice of `phenix.model_vs_data` as the headline scorer in the main figure.

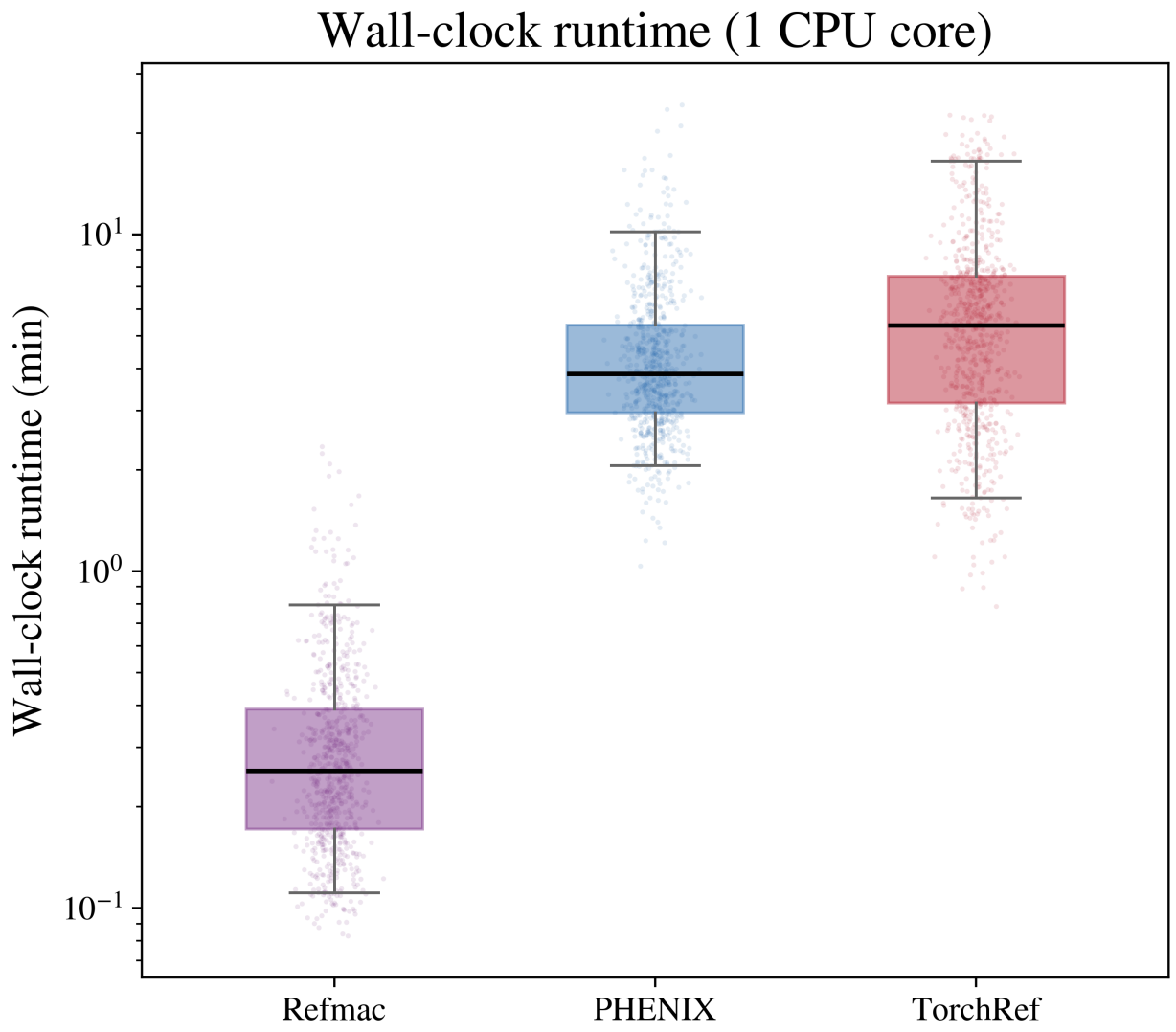


Figure S4: Single-core wall-clock runtime on the AlphaFold benchmark, with every engine pinned to a single CPU thread (TorchRef `TORCHREF_NUM_THREADS=1`; `phenix.refine refinement.main.nproc=1`; REFMAC5 with `OMP_NUM_THREADS=1`). The same conserved cohort as in main-text Fig. 2C was re-timed at one core ($n = 723$). Box plots are 25–75th percentile with 5–95th percentile whiskers; the faint jittered strip shows individual structures. Ratios of the engine medians show TorchRef is $\approx 21.0\times$ slower than REFMAC5 and $\approx 1.4\times$ slower than `phenix.refine` at one core; this contrasts with the four-core comparison in Fig. 2C, where TorchRef pulls clearly ahead of `phenix.refine`. TorchRef also shows a noticeably wider per-structure runtime distribution than `phenix.refine` and REFMAC5, with a long upper tail driven by the largest structures in the cohort. REFMAC5's single-core performance is essentially unchanged from its four-core performance because it does not use OpenMP parallelism.

Supplementary methods

1 Scope and notation

This document describes the quantities TorchRef refines, the structure factor calculation, and the implemented target functions. Where a default value is sourced from external data (e.g. the CCP4 monomer library) and varies per restraint, this is noted explicitly.

Throughout, we use the following notation. F refers to the (complex) structure factor and $|F|$ the amplitude. F_o is the observed structure factor amplitude and $F_c = |F_{\mathrm{calc}}|$ the model amplitude; σ the experimental uncertainty on F_o. The scattering vector is $\mathbf{s}$ with $s = |\mathbf{s}| = 2\sin\theta/\lambda$ (Å^{-1}); $d^* = 1/d = s$ and $d^{*2} = 1/d^2 = s^2$; $s_{1/2} = \sin\theta/\lambda = s/2$ (Å^{-1}) is used when atomic form factors are tabulated in the $(\sin\theta/\lambda)^2$ convention of the *International Tables for Crystallography*, Vol. C (ITC92) (Wilson, 1992). The reflection multiplicity ε is the number of rotational symmetry operations mapping $\mathbf{h} \to \pm\mathbf{h}$, including the inversion $\mathbf{h} \to -\mathbf{h}$. This Friedel-aware convention differs from the classical Wilson ε (which counts only $R^{\mathsf{T}}\mathbf{h} = \mathbf{h}$) by inclusion of inversion symmetry. B denotes an isotropic atomic displacement parameter (Å^2) and $\mathbf{U}$ an anisotropic ADP tensor (Å^2). For an isotropic atom $U_{\mathrm{iso}} = B/(8\pi^2)$; the anisotropic Debye–Waller factor uses $\exp(-2\pi^2\mathbf{s}^{\mathsf{T}}\mathbf{U}\mathbf{s})$, where $\mathbf{s}$ is the Cartesian scattering vector (see Eq. S15), with equivalent isotropic $B_{\mathrm{eq}} = (8\pi^2/3)\,\mathrm{tr}(\mathbf{U})$. I_0, I_1 are modified Bessel functions of the first kind, and $\mathrm{i0e}(x) = e^{-x}I_0(x)$, $\mathrm{i1e}(x) = e^{-x}I_1(x)$ denote their exponentially scaled forms, and $\mathrm{logi0e}(x) = \log\mathrm{i0e}(x) = \log I_0(x) - x$ its logarithm. We write $\mathbf{h}$ for a Miller-index triple; in reflection sums and subscripts of the form $\sum_h$ or w_h, the symbol h is the running enumeration index over the flattened reflection list.

2 Reflection data preparation

Before refinement begins, observed intensities are converted to amplitudes, reflections are flagged for cross-validation, and outliers and suspicious uncertainties are masked. All steps operate on a `ReflectionData` object that stores reflections in the standard CCP4 asymmetric-unit (ASU) convention, which all downstream HKL operations assume.

2.1 Intensity-to-amplitude conversion (French–Wilson)

When the input data are intensities I with uncertainties σ_I, TorchRef converts them to amplitudes via the French–Wilson treatment (French & Wilson, 1978). The implementation follows the `mmtbx` French–Wilson routine, with separate centric and acentric branches.

Reflections are first binned in equal-count resolution shells (60 bins, minimum 40 reflections per bin). The per-shell Wilson mean $\langle I\rangle$ is estimated on the available reflections in each shell, then linearly interpolated in d-spacing to each reflection. Given the French–Wilson normalised intensity $h = I/\sigma_I - \sigma_I/\langle I\rangle$ (acentric; the centric branch uses $\sigma_I/2\langle I\rangle$ for the second term), the conditional posterior mean and variance of $|F|$ are evaluated and scaled by $\sqrt{\sigma_I}$. Two regimes are used: a

precomputed lookup table for small h (acentric $h < 3$, centric $h < 4$), and a closed-form asymptotic expansion at large h using the centric and acentric series. Reflections with non-finite I or σ_I are filled with NaN and excluded from downstream computation; strongly negative or very-low-σ reflections within the standard French–Wilson regime are *not* excluded but have h clamped to the boundary lookup value $h_{\min} = -4.0$; a companion threshold $(I/\sigma)_{\min} = -3.7$ marks the boundary of the separate boolean `valid_mask`. The per-shell Wilson mean is recomputed on every forward pass, so the conversion tracks updated intensities without re-preprocessing.

2.2 Outlier and suspicious-σ flagging

All masking is model-free and applied once at data load: no calculated-amplitude outlier rejection is performed.

Data-validity flagging. Reflections with a non-finite F or σ, or a non-positive F, are dropped (mask `sanity_F`). Intensity inputs additionally pass the French–Wilson input guard, which drops measurements too negative to be a noisy sample of any non-negative true intensity, the case for which its posterior integral cannot be evaluated. With $\langle I \rangle$ the mean intensity of the reflection's shell (estimated per shell and interpolated in resolution), the guard keeps a reflection when

$$h \;=\; \frac{I}{\sigma_I} - \frac{\sigma_I}{\varepsilon_{\mathrm{W}}\,\langle I \rangle} \;\geq\; -4, \qquad \varepsilon_{\mathrm{W}} = \begin{cases} 2 & \text{centric,} \\ 1 & \text{acentric,} \end{cases} \tag{S1}$$

(together with $I/\sigma_I \geq -3.7$), h being the argument of the Wilson prior folded with Gaussian measurement error, the centric case carrying twice the prior variance. This is a validity guard on the intensity-to-amplitude conversion, distinct from the outlier test below.

Wilson-outlier flagging. The outlier test is model-free (it never sees an F_{calc}) and asks how improbable each observation is under Wilson statistics for its resolution shell, multiplicity and centricity, in *both* tails. Its reference is the Wilson scale $S = \varepsilon_{\mathrm{W}}\,\Sigma$ (the expected intensity), where Σ is estimated from the data as a robust per-shell median (the median of Exp(1) is $\ln 2$ and of χ^2_1 is 0.4549, converting a shell median back to Σ with no fit), taken separately for centric and acentric reflections, scaled by the reflection multiplicity, and given an anisotropic correction $\Sigma_{\mathrm{iso}}(d)\,\exp(-2\pi^2\,\mathbf{s}^{\top}\mathbf{U}\,\mathbf{s})$ fitted to the observations: the median stops a strong outlier from inflating its own reference, and the anisotropy stops an isotropic Σ from flagging the strong reciprocal direction of every anisotropic dataset. Convolving the Wilson prior with the Gaussian measurement error gives a predictive density whose upper tail is closed form,

$$P(I' > I) \;=\; Q\Big(\tfrac{I}{\sigma_I}\Big) + \exp\!\left(\frac{\sigma_I^2}{2S^2} - \frac{I}{S}\right)\Phi\Big(\tfrac{I}{\sigma_I} - \tfrac{\sigma_I}{S}\Big), \qquad Q = 1 - \Phi, \tag{S2}$$

whose Φ argument is the statistic of Eq. S1 evaluated against this robust S; the closed form is exact for acentric reflections, with centrics handled by the same expression under the French–Wilson factor-of-two convention ($S = 2\Sigma$), which captures the correct tail exponent. A reflection is rejected when it sits further into the strong tail ($\log P(I' > I)$ below threshold: zingers, ice rings, mis-integrated spots) or the weak tail (h below threshold) than the dataset size can account for: the threshold is a family-wise error rate α/N over the N tested reflections (default $\alpha = 0.01$), so it tightens as the dataset grows rather than rejecting a fixed fraction. Only reflections finer than 4 Å are tested, where bulk solvent no longer dominates and the shells are well populated, and Σ and $\mathbf{U}$ are refit in a second pass with the first pass's outliers held out. Amplitude-only inputs reconstruct intensities ($I = F^2$, $\sigma_I = 2F\sigma_F$) first; the negative tail is then blind (the amplitude is a positive posterior mean) but the strong tail, where outliers sit, is unaffected.

Suspicious-σ flagging. A secondary, diagnostic-only pass, superseded in the automatic load path by the Wilson test above. Reflections with non-finite σ, and those whose $\log\sigma$ lies more than $z = 5$ standard deviations from the bulk distribution, are flagged (mask `flagged_sigma`); the standard deviation is stabilised with a small term proportional to the mean ($10^{-5}\langle\log\sigma\rangle$).

2.3 Cross-validation flags

TorchRef supports a three-way *work / free / validation* partition of the reflections. The work and free sets follow the cross-validation recipe of (Brünger, 1992): free reflections do not contribute direct coordinate or ADP gradients and serve two roles. First, they provide the reference for the R_{free} profile. Second, they are the sample from which the model-error component β_{model} and total variance β of the σ_A maximum-likelihood target (see *X-ray target functions → Maximum-likelihood σ_A target* below) are derived, via a maximum-likelihood estimate of σ_A, using noise-corrected moments on equal-count free-set shells. Because this detached error model uses the free reflections, R_{free} is a useful profile diagnostic but not a strictly untouched held-out score.

R-free flags are assigned by resolution-stratified sampling: equal-count shells ordered in d^{*2} (10 bins, minimum 1000 reflections per bin) with a target free fraction of 2% per bin. A separate per-bin floor of 50 free reflections (clamped to the bin size on tiny bins) keeps the per-shell statistics usable for small datasets. The draw is over unique CCP4-asymmetric-unit reflections rather than over rows: the two members of a Bijvoet pair share a canonical ASU index and are assigned the same flag, so Friedel mates are never split across the work/free boundary.

The optional *validation set* is a disjoint third subset carved from the free set on request (default: half of the free reflections), using the same resolution-stratified sampler and stored in a separate boolean `validation_flags` buffer that survives MTZ round-trip; the binary R-free flags are unchanged. Standard refinement does not use it: the optimisation uses the work set and R_{free} is reported against the free set as usual. The validation set supports tuning choices outside the likelihood (model size, regularisation strength, ensemble size) without contaminating the free-set statistic.

3 Parameterisation

A central design choice of TorchRef is to map atomic displacement parameters (ADPs) and occupancies to unconstrained latent parameters through an invertible transform: a log map for the isotropic B, a Cholesky factor for the anisotropic $\mathbf{U}$, and a logit for the occupancy. The mapping enforces the physical parameter bounds explicitly: negative B, indefinite $\mathbf{U}$ and occupancy outside $[0, 1]$ are impossible by construction, so an unconstrained optimiser can act on every parameter type without bound constraints. The production schedule retains the tested blockwise protocol, alternating scaler, xyz, and ADP/occupancy blocks per macrocycle (see *Macrocycle structure* below). All gradients with respect to the latent variables are obtained by automatic differentiation. Each parameterisation below is an independent PyTorch module behind a uniform interface, so any can be swapped for a user-defined alternative without touching the rest of the pipeline.

3.1 Coordinates

Atomic coordinates are parameterised in Cartesian space, $\mathbf{x}_{\text{cart}} \in \mathbb{R}^{N\times 3}$ (Å). Fractional coordinates are derived on demand from the orthogonalisation matrix $\mathbf{O}$ via Eq. S3:

$$\mathbf{x}_{\text{frac}} = \mathbf{x}_{\text{cart}}\,\mathbf{O}^{-\mathsf{T}}, \qquad \mathbf{x}_{\text{cart}} = \mathbf{x}_{\text{frac}}\,\mathbf{O}^{\mathsf{T}}. \tag{S3}$$

Internal/torsion-angle parameterisations have been investigated in the past but have since been deprecated.

3.2 Atomic displacement parameters

Isotropic B-values are refined in log-space, guaranteeing positivity for any real latent p (Eq. S4):

$$B = \exp(p), \qquad p = \log(\max(B, \epsilon)), \qquad \epsilon = 0.1\ \text{Å}^2. \tag{S4}$$

Anisotropic ADPs are parameterised through a Cholesky factor $\mathbf{U} = \mathbf{L}\mathbf{L}^{\mathsf{T}}$ with six free parameters, which keeps $\mathbf{U}$ positive-definite by construction, with $\mathbf{L}$ as in Eq. S5:

$$\mathbf{L} = \begin{pmatrix} \exp(p_0) + \epsilon & 0 & 0 \\ p_3 & \exp(p_1) + \epsilon & 0 \\ p_4 & p_5 & \exp(p_2) + \epsilon \end{pmatrix}, \qquad \epsilon \text{ a small positive floor.} \tag{S5}$$

The decomposition is performed only at initialisation (and on freeze/unfreeze), never inside the forward loop, so the optimiser can never drive $\mathbf{U}$ indefinite and produce NaNs in the structure factor FFT.

3.3 Occupancies

Occupancies are refined through a logit/sigmoid map (Eq. S6),

$$\mathrm{occ} = \mathrm{sigm}(p) = \frac{1}{1 + e^{-p}}, \tag{S6}$$

with initial values clamped just inside $[0, 1]$. Shared sites collapse to a single latent via an expansion mask. Linked occupancies (e.g. alternative conformations that must sum to unity within an alt-conf group) are enforced by parameterisation: the effective occupancy entering the structure factor calculation is the group-normalised sigmoid of the logit, as in Eq. S7.

$$\mathrm{occ}_i \;=\; \frac{\mathrm{sigm}(p_i)}{\sum_{j \in \mathrm{group}} \mathrm{sigm}(p_j)}. \tag{S7}$$

3.4 Refinable / fixed parameter split

Each refinable parameter type keeps the full atom set as a frozen buffer and an index that maps the currently-refinable subset into it. The full parameter array is reconstructed at every forward pass by scattering the refinable subset into that buffer; freezing or unfreezing a selection rebuilds the index.

3.5 ADP-mode switching at runtime

Atomic displacement parameters can be switched between isotropic and anisotropic storage at runtime, without rebuilding the model. A mixed isotropic + anisotropic representation is realised by a per-atom `mmtbx`-style selection string (default: `"not resname HOH and not element H"`) that marks the atoms carrying anisotropic ADPs. The conversion math:

- Anisotropic $\rightarrow$ isotropic: the equivalent isotropic B-value is taken as the standard equivalent of the diagonal trace, $B_{\mathrm{eq}} = (8\pi^2/3)\,\mathrm{tr}(\mathbf{U})$.
- Isotropic $\rightarrow$ anisotropic: the isotropic B is broadened into a diagonal $\mathbf{U}$ with $U_{\mathrm{iso}} = B/(8\pi^2)$ and zero off-diagonal components. The Cholesky parameterisation (see "Anisotropic ADPs" above) is rebuilt from the new $\mathbf{U}$.

The repartition updates all downstream caches (structure factor indexing, refinable masks, and PDB anisotropy flags); it is a full repartition, not a value-only conversion.

4 Structure factor calculation

TorchRef provides two structure factor back-ends: a direct (Debye) summation and an FFT-based path, the latter being the default for production refinement. The sign convention is $\exp(+2\pi i\,\mathbf{h} \cdot \mathbf{r})$.

4.1 Reciprocal basis and late symmetry expansion

The reciprocal basis is built from the cell parameters $(a, b, c, \alpha, \beta, \gamma)$ in a lower-triangular real-space convention, as in Eq. S8,

$$\mathbf{a}^* = \frac{\mathbf{b} \times \mathbf{c}}{V}, \quad \mathbf{b}^* = \frac{\mathbf{c} \times \mathbf{a}}{V}, \quad \mathbf{c}^* = \frac{\mathbf{a} \times \mathbf{b}}{V}, \quad V = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}), \tag{S8}$$

with $\mathbf{s} = h\,\mathbf{a}^* + k\,\mathbf{b}^* + l\,\mathbf{c}^*$.

The density and FFT are computed in space group $P1$, and crystal symmetry is applied in reciprocal space (Eq. S9), which is exact on the integer grid and avoids expanding symmetry-related copies of every atom,

$$F_{\text{sym}}(\mathbf{h}) = \sum_{\text{ops}} \exp(2\pi i\,\mathbf{h} \cdot \mathbf{t}_{\text{op}})\, F_{P1}\left(\mathbf{R}_{\text{op}}^{\mathsf{T}}\mathbf{h}\right). \tag{S9}$$

The grid is chosen to be compatible with the space group, and optimised to be FFT-friendly.

Reflections are canonicalised to the CCP4 asymmetric unit by `gemmi.ReciprocalAsu`, via a vectorised fast path that covers the ten distinct CCP4 ASU conditions (sufficient for all 230 space groups (Agirre *et al.*, 2023; Wojdyr, 2022)). Canonicalisation also applies the phase correction $\varphi_{\text{new}} = \varphi_{\text{old}} + \varphi_{\text{shift}}$, with a sign flip on Friedel mates, so that ASU-stored amplitudes and phases remain consistent with the unit-cell origin.

4.2 Atomic scattering factors

Atomic form factors use the ITC92 parameterisation, which is a sum of four Gaussians plus a constant (Wilson, 1992) as in Eq. S10,

$$f(s) = \sum_{i=1}^{4} a_i \exp\left(-b_i\, s^2/4\right) + c. \tag{S10}$$

For vectorisation, TorchRef packs the constant c as a fifth Gaussian ($a_5 = c$, $b_5 = 0$), so evaluation is a single $(N_{\text{atoms}}, 5)$ contraction.

The form factors are shared by both structure factor back-ends: a Z-based lookup against a precomputed table of neutral atoms ($Z = 1 \ldots 103$) yields the $(N_{\text{atoms}}, 5)$ coefficient tensors $\mathbf{A}$ and $\mathbf{B}$. The production path uses the neutral form factor regardless of the input PDB's formal-charge column; ion-specific entries are available only through an explicit helper.

4.3 FFT path and electron-density sampling

In the FFT path the model electron density is sampled onto a real-space grid with spacing $d_{\min}/3$, and structure factors are obtained by an inverse FFT, as in Eq. S11,

$$F(\mathbf{h}) = \frac{V_{\text{cell}}}{N_{\text{voxels}}} \,\text{ifftn}(\rho), \tag{S11}$$

where `ifftn` denotes the un-normalised inverse FFT (`torch.fft.ifftn(..., norm="forward")`); the prefactor $V_{\text{cell}}/N_{\text{voxels}}$ is the voxel volume that converts the discrete sum into the continuum integral, yielding complex structure factors on the absolute scale.

Each atom's density is a sum of 3D Gaussians (one per scattering term) centred on the atomic position. TorchRef truncates it at a per-atom radius large enough to sample the atom's widest Gaussian out to N_σ standard deviations (default $N_\sigma = 3$), rounded up to the nearest 0.25 Å and clamped to $[2, 7]$ Å:

$$r_{\text{atom}} = \text{clamp}\big(\text{ceil}_{0.25}(N_\sigma\, \sigma_{\text{eff}}),\ [2.0,\ 7.0]\ \text{Å}\big), \tag{S12}$$

where $\sigma_{\text{eff}} = \sqrt{(b_{\text{form}} + B)/(8\pi^2)}$ is the effective real-space width built from the broadest ITC92 Gaussian width b_{form} and the isotropic ADP B; anisotropic atoms use the largest principal axis of the ADP tensor, $\sigma^2_{\text{eff}} = b_{\text{form}}/(8\pi^2) + \lambda_{\max}(\mathbf{U})$. Expressing the cutoff in sigmas gives every atom the same fractional Gaussian tail, so the truncation residual is set by the single knob N_σ rather than by the worst atom; the default $N_\sigma = 3$ was validated against the direct-summation reference.

The two device paths implement this per-atom radius differently. The CPU sampler buckets atoms by box radius so that same-size atoms share a single voxel loop. The GPU path runs one Triton program per atom, each sizing its bounding box from r_{atom} and truncating to the per-atom sphere $r^2 \le r^2_{\text{atom}}$.

TorchRef exploits the separability of a Gaussian on an orthogonal grid. Writing the local coordinate offset as $(\delta x, \delta y, \delta z)$ along the three orthogonal Cartesian axes, an isotropic Gaussian of width w factorises as in Eq. S13,

$$\exp\left(-\frac{\delta x^2 + \delta y^2 + \delta z^2}{2w^2}\right) = \exp\left(-\frac{\delta x^2}{2w^2}\right)\exp\left(-\frac{\delta y^2}{2w^2}\right)\exp\left(-\frac{\delta z^2}{2w^2}\right), \tag{S13}$$

so the 3D contribution is the outer product of three 1D profiles. For an orthogonal cell this reduces the per-atom cost from a naive $(2R+1)^3$ exponential evaluations to $3(2R+1)$ per Gaussian term. For monoclinic and hexagonal cells, exactly one pair of axes is oblique, so the density factorises into one 1D profile along the orthogonal axis times a 2D correction evaluated on the oblique plane, at a per-atom cost of $(2R+1) + (2R+1)^2$ evaluations per term. For triclinic cells, or for anisotropic scattering where no axis decouples, TorchRef falls back to the full 3D evaluation.

The FFT path is validated against the eager direct-summation reference using cosine-similarity and gradient-norm-ratio metrics on a scalar loss.

4.4 Direct summation

For isotropic ADPs, the structure factor is computed by direct summation as in Eq. S14,

$$F(\mathbf{h}) = \sum_j \mathrm{occ}_j \, f_{0,j}(s) \, \exp\!\left(-B_j \, s^2/4\right) \exp(2\pi i \, \mathbf{h} \cdot \mathbf{x}_j) \,, \tag{S14}$$

with $\mathbf{x}_j$ the fractional position of atom j (so $\mathbf{h} \cdot \mathbf{x}_j$ is dimensionless), using the thermally attenuated neutral form factor $f_{0,j}(s)$. The anomalous contribution is not part of this summation; as on the FFT path, it is added as a separate post-processing step on the backend structure-factor output, as described in Section 4.5 (in the current release wired only to the FFT path). For atoms refined with anisotropic ADPs, the isotropic Debye–Waller factor $\exp(-B_j s^2/4)$ in Eq. S14 is replaced atom-wise by the anisotropic form in Eq. S15; a mixed model with both iso- and aniso-refined atoms is the sum of the two corresponding sub-sums,

$$\exp\!\left(-2\pi^2 \, \mathbf{s}^{\mathsf{T}} \mathbf{U}_j \, \mathbf{s}\right) . \tag{S15}$$

Summation is auto-batched and checkpointed over reflections to respect the memory budget. The eager direct-summation forward serves as the autograd reference for the structure factor calculation; the GPU Triton kernels and the gradient-checkpointed CPU path are validated against it.

4.5 Anomalous scattering

The dispersive correction f' is applied whenever the model contains significant anomalous scatterers, and preserves Friedel's law; the absorptive term f'', which breaks Friedel's law and produces Bijvoet differences, is included only when the reflection data are loaded as unmerged Friedel pairs. The atomic form factor is augmented with the dispersive and absorptive corrections of Eq. S16,

$$f(s, \lambda) \;=\; f_0(s) + f'(\lambda) + i \, f''(\lambda), \tag{S16}$$

with f' and f'' obtained per element from the Cromer–Liberman calculation (Cromer & Liberman, 1981) via `gemmi`, evaluated at the model wavelength (default 1 Å). The lookup is wavelength-dependent and per-element but does not modulate with resolution; energy is converted from the wavelength using $E_{\mathrm{eV}} = hc/\lambda$.

For per-step efficiency only elements with non-negligible $|f'|$ or $|f''|$ (default threshold $0.5\,e$) contribute. Light atoms (C, N, O, H) are automatically filtered out at this stage; per-element correction values and atom-index tensors are cached on the model.

The anomalous-correction cache is keyed on the model's element list, which fixes $f'(\lambda)$ and $f''(\lambda)$ at the construction wavelength, and is rebuilt when that list changes (e.g. after adding or removing heavy atoms); a wavelength change requires an explicit cache reset.

No Debye–Waller term is applied to the anomalous heavy-atom contribution; for floppy heavy

atoms at high resolution, this is a known approximation.

4.6 Bulk-solvent mask

The bulk-solvent mask is built in an `mmtbx`-style three-step procedure (Afonine *et al.*, 2005): (i) atom-centred dilation by the van der Waals radius of each heavy atom plus a probe radius R_{probe} (default 1.1 Å); (ii) symmetry expansion of the ASU mask to $P1$ on the density grid; (iii) erosion by a shrinkage radius R_{shrink} (default 0.9 Å), implemented as a solvent-side dilation followed by intersection with the mask complement. All distances respect the periodic boundary via the minimum-image rule in fractional coordinates, which handles atoms that straddle cell faces.

$F_{\mathrm{mask}}(\mathbf{h})$ is obtained from a single inverse FFT of the binary solvent mask and scaled (Eq. S22) to produce the bulk-solvent contribution $F_{\mathrm{sol}}(\mathbf{h})$.

The solvent parameters $\log k_{\mathrm{sol}}$, the falloff half-point $\log s_h^2$ and sharpness $\log n$, and an optional phase offset $\varphi_{\mathrm{offset}}$ are `nn.Parameter`s on the solvent module and are refined alongside the other scaling parameters (Section 4.7). The default scale follows Afonine *et al.* ($k_{\mathrm{sol}} = 0.35$); the falloff defaults to $n = 5$, so the solvent term switches off sharply near the half-point rather than as a Debye–Waller factor (the $n = 1$ special case), with s_h^2 initialised from a chosen resolution d_h via $s_h^2 = 1/(4d_h^2)$.

4.7 Scaling

Model amplitudes are placed on the scale of the data through Eq. S17,

$$F_{\mathrm{scaled}}(\mathbf{h}) = K_{\mathrm{overall}}(\mathbf{h})\, b_{\mathrm{overall}}(\mathbf{h})\, \big(\mathrm{aniso}(\mathbf{h})\, F_{\mathrm{calc}}(\mathbf{h}) + F_{\mathrm{sol}}(\mathbf{h})\big), \tag{S17}$$

with the per-factor expressions given by Eq. S22,

$$K_{\mathrm{overall}}(\mathbf{h}) = \exp\Big(\mathbf{d}(\mathbf{h})^{\mathsf{T}}\mathbf{c}_{\mathrm{iso}}\Big) \tag{S18}$$

$$b_{\mathrm{overall}}(\mathbf{h}) = \exp\big(-B_{\mathrm{bin}}[\mathrm{bin}(\mathbf{h})]\, s^2/4\big) \tag{S19}$$

$$\mathrm{aniso}(\mathbf{h}) = \exp\Big(-2\pi^2\, \mathbf{s}^{\mathsf{T}}\mathbf{U}_{\mathrm{scaler}}\, \mathbf{s}\Big) \tag{S20}$$

$$F_{\mathrm{sol}}(\mathbf{h}) = k_{\mathrm{sol}}\, e^{\,i\varphi_{\mathrm{offset}}}\, \exp\Big(-\ln 2\, \big(s_{1/2}^2/s_h^2\big)^n\Big)\, F_{\mathrm{mask}}(\mathbf{h}) \tag{S21}$$

$$k_{\mathrm{sol}} = \exp(\mathrm{log_k_solvent}) \tag{S22}$$

The isotropic scale $\mathbf{d}(\mathbf{h})^{\mathsf{T}}\mathbf{c}_{\mathrm{iso}}$ is a smooth Chebyshev polynomial in $|\mathbf{s}|$ evaluated per reflection, with the design row $\mathbf{d}(\mathbf{h})$ holding the Chebyshev basis and $\mathbf{c}_{\mathrm{iso}}$ the fitted coefficients, clamped per reflection to $[-10, 10]$; it is not a per-shell step function. Equal-count resolution shells (10 by default, 20 for the standalone `Scaler`, and reduced for sparse data) are used only to seed the initial coefficients. The per-bin b_{overall} factor is an *optional* component (off by default, enabled by `setup_bin_wise_bfactor()`); when disabled, $b_{\mathrm{overall}} \equiv 1$. The set of scaler parameters fitted by L-

BFGS in the default configuration ($\mathbf{c}_{\text{iso}}$, $\mathbf{U}_{\text{scaler}}$, $\log k_{\text{sol}}$, $\log s_h^2$, $\log n$, and φ_{offset}) is optimised against a separate, α-free scale objective. The default is unit-weight least squares `ls`; the σ_{obs}-weighted Gaussian `nll` and the $\alpha \equiv 1$ Read likelihood `ml_noalpha` are the other permitted choices.

A single restraint keeps the anisotropic scaler from soaking up information that should remain in the atomic model.

Anisotropic-scale pin. The anisotropic scaler $\mathbf{U}_{\text{scaler}}$ can absorb a B-factor correction that should remain in the atomic ADPs. A complementary target penalises the magnitude of $\mathbf{U}_{\text{scaler}}$ through the sum of squares of its components, $\mathcal{L}_U = \left(\sum \mathbf{U}_{\text{scaler}}^2\right) / \sum_{\text{work}} |F_{\text{obs}}|^2$, using the same $1/\sum_{\text{work}} |F_{\text{obs}}|^2$ normalisation as the scale objective so the two are directly comparable. This shrinks $\mathbf{U}_{\text{scaler}}$ toward zero, keeping the anisotropic correction minimal so it captures only what the data genuinely require rather than ADP information that belongs in the atomic model. The overall isotropic B-trend is instead re-absorbed by the scaler at the start of each macrocycle (see *Macrocycle structure*), so the ADP–overall-B degeneracy is managed through the refinement schedule rather than a restraint.

Scaling is implemented in a model-aware object that computes F_{calc} internally; a model-independent variant taking F_{calc} as input underlies the multi-dataset `CollectionScaler` (Section 9).

5 X-ray target functions

All X-ray targets operate on the work/free split of the data and recompute F_{calc} fresh from the live model on every evaluation. The factory exposes seven modes drawn from only *three* underlying likelihood shapes: a Gaussian on the amplitude, the Read maximum-likelihood (Rice / folded-normal) function, and its measurement-error-marginalised full form. They differ only in the choice of (distribution) × (variance) × (conditional mean), as summarised in Table S1. The default is `ml`, the Read MLF centred on the Luzzati mean $\alpha|F_c|$. The X-ray target's autograd path is

mode	distribution	conditional mean	variance
`ml` (default)	Rice / folded normal	$\alpha\|F_c\|$	$\varepsilon\beta$
`ml_noalpha`	Rice / folded normal	$\|F_c\|$	$\varepsilon\beta$
`ml_full`	Rice $\otimes$ Gaussian, marg.	$\alpha\|F_c\|$	$\varepsilon\beta_{\text{model}} + \sigma^2$
`nll_beta`	Gaussian on $\|F\|$	$\|F_c\|$	$\varepsilon\beta \to$ amplitude
`nll`	Gaussian on $\|F\|$	$\|F_c\|$	σ^2
`ls`	least squares	$\|F_c\|$	unit weights
`ls_wunit_k1`	least squares	$\|F_c\|$	unit, global k

Table S1: The X-ray target taxonomy. α is the Luzzati mean coupling and β the per-shell total error variance used directly by `ml`, `ml_noalpha`, and `nll_beta`. `ml_full` instead uses $\varepsilon\beta_{\text{model}}$ for the complex model error and marginalises the experimental amplitude error separately. The model-error component is derived from the free-set σ_A, whereas σ is experimental input (see *Maximum-likelihood targets*). The `nll` and least-squares rows do not estimate σ_A. Here ε is the Friedel-aware multiplicity and σ the experimental amplitude e.s.d.

verified by `torch.autograd.gradcheck` in double precision against central-difference numerical gradients.

5.1 Least-squares and Gaussian targets

The `ls` target minimises the unit-weighted sum of squared amplitude residuals, Eq. S23:

$$L = \sum_h \tfrac{1}{2} \left(F_o - |F_c|\right)^2, \tag{S23}$$

with the per-bin scale k folded into F_c by the upstream `Scaler` (Section 4.7). An inverse-variance weighting ($w_h = 1/\sigma_{\text{safe}}^2$, $\sigma_{\text{safe}} = \max(\sigma, 0.1 \cdot \text{median}\,\sigma)$) is selectable but, up to a parameter-independent constant, has the same gradients as `nll` (below), so it reproduces the `nll` trajectory rather than acting as a distinct target. The `ls_wunit_k1` variant keeps unit weights but recomputes a detached global work-set scale each evaluation (the `mmtbx` `wunit` convention) and applies it to the amplitude supplied by the upstream calculation. In the rigid-body path that upstream calculation retains the one-bin scaler and bulk-solvent contribution described in Section 11.3.

The Gaussian amplitude target `nll` adds the Gaussian log-normalisation terms,

$$L = \sum_h \left[\tfrac{1}{2} \left(\frac{F_o - |F_c|}{\sigma_{\text{safe}}} \right)^2 + \log \sigma_{\text{safe}} + \tfrac{1}{2} \log(2\pi) \right]. \tag{S24}$$

Because σ_{safe} is a fixed data quantity, these extra terms are constant in the model parameters and do not affect the gradient. Weighted by the experimental e.s.d. alone, `nll` carries no model-error term and behaves like ordinary least squares; it is provided for comparison and is not recommended for refinement, since the observed e.s.d.s over-weight the fit to the data. A companion mode `nll_beta` keeps the same Gaussian shape but replaces the measurement-only variance with the σ_A-derived total variance $\varepsilon\beta$ defined below (converted to an amplitude variance); it is the large-signal Gaussian limit of the $\alpha = 1$ `ml_noalpha` likelihood and serves as a diagnostic that isolates the variance model from the Rice shape.

5.2 Maximum-likelihood targets: the Read / Rice family

The default `ml` target uses the Read maximum-likelihood function (Read, 1986): a Rice distribution for acentric reflections and its folded-normal limit for centrics, with complex variance $\Sigma = \varepsilon\beta$ and conditional mean $\alpha|F_c|$, where α is the Luzzati mean coupling (the D-factor for unnormalised amplitudes) and β the total variance in F^2 units, comprising the model-error term β_{model} and mean measurement variance S^2 (a perfect model has $\beta_{\text{model}} \to 0$, not generally $\beta \to 0$). Writing

the coupled model amplitude as αF_c, the per-reflection negative log-likelihood is Eq. S26:

$$L_{\text{acentric}} = -\log \frac{2F_o}{\Sigma} + \frac{F_o^2}{\Sigma} + \frac{\alpha^2 F_c^2}{\Sigma} - \left[\text{logi0e}\left(\frac{2F_o\,\alpha F_c}{\Sigma}\right) + \frac{2F_o\,\alpha F_c}{\Sigma}\right], \tag{S25}$$

$$L_{\text{centric}} = -\frac{1}{2}\log \frac{2}{\pi\Sigma} + \frac{F_o^2}{2\Sigma} + \frac{\alpha^2 F_c^2}{2\Sigma} - \frac{F_o\,\alpha F_c}{\Sigma} - \log \frac{1 + e^{-2F_o\,\alpha F_c/\Sigma}}{2}, \tag{S26}$$

with ε the Friedel-aware multiplicity defined in *Scope and notation*. The mean coupling enters the likelihood only through the product αF_c (the model terms carry $\alpha^2 F_c^2$ and $2F_o\,\alpha F_c$), so applying it is exact. The three family members differ in this mean and in their treatment of the measurement-error component: `ml` (the default) centres on $\alpha|F_c|$; `ml_noalpha` pins the coupling to 1 (mean $|F_c|$), the correct choice for a *scale* fit; and `ml_full` centres on $\alpha|F_c|$ but marginalises the measurement error explicitly (both below). Formally, when the model carries no information, $\sigma_A \to 0$ and $\alpha \to 0$, so the non-centrality vanishes and the Rice distribution approaches the Wilson prior; the numerical backstop $\alpha \geq 0.1$ (below) keeps the estimator from reaching that limit.

Estimating the coupling and the variance. Both α and the model-error component β_{model} (hence total β) derive from a single per-shell correlation $\sigma_A \in [0, 1]$, estimated by maximum likelihood on the free set. Here $\langle x \rangle$ denotes the implementation's weighted shell average, $\sum_h w_h x_h / \sum_h w_h$, with $w_h = 1$ for centric and $w_h = 2$ for acentric reflections (Lunin–Skovoroda weighting (Lunin & Skovoroda, 1995)). Define the observed second moment, mean measurement variance, noise-corrected data power, and model power as

$$B = \left\langle \frac{F_o^2}{\varepsilon} \right\rangle, \qquad S^2 = \left\langle \frac{\sigma_F^2}{\varepsilon} \right\rangle, \qquad \Sigma_N = B - S^2, \qquad \Sigma_P = \left\langle \frac{F_c^2}{\varepsilon} \right\rangle. \tag{S27}$$

The implementation then derives

$$\alpha = \sigma_A \sqrt{\frac{\Sigma_N}{\Sigma_P}}, \qquad \beta_{\text{model}} = (1 - \sigma_A^2)\Sigma_N, \qquad \beta = \beta_{\text{model}} + S^2, \qquad \Sigma = \varepsilon\beta. \tag{S28}$$

Thus $\sigma_A = \sqrt{1 - \beta_{\text{model}}/\Sigma_N}$; the measurement variance S^2 is not included inside this correlation identity. For the standard `ml` target the complex variance is Σ, Eq. S29:

$$\Sigma = \varepsilon\,\beta. \tag{S29}$$

The estimator evaluates the joint α–β likelihood while fitting the single free quantity σ_A for every model-error family member, including those (`ml_noalpha`, `nll_beta`) whose refinement likelihood then pins the mean at $|F_c|$, because pinning the mean during estimation biases σ_A high. Because α, β_{model}, and β are algebraic consequences of the fitted σ_A, the shell second-moment identity

$$\alpha^2\Sigma_P + \beta_{\text{model}} + S^2 = B \tag{S30}$$

holds exactly before numerical backstops are applied. This is a shell-averaged statement, not a pointwise constraint on any individual reflection.

Decoupling the scale from the coupling. Because α and the overall scale k enter the likelihood only through the product αk, an α-centred target would let the scale absorb $1/\alpha$ and inflate every R-factor computed from $k|F_c|$. The scale is therefore fit against the α-free targets only (`ls`, `nll` and `ml_noalpha`), fixing the magnitude gauge by an α-independent criterion before α enters the refinement target. `ml_full` instead centres on $\alpha|F_c|$ but marginalises the unknown error-free amplitude, so the measurement error enters as an amplitude-only Gaussian while β_{model} keeps the complex (phase) component it physically has. This is the treatment preferred when measurement errors are large, at $\sim 4\times$ the per-gradient cost.

The estimator. σ_A (hence α and β) is estimated per equal-count free-set shell by maximum likelihood. It is detached (no autograd path runs from β to the atomic parameters) and held fixed within each L-BFGS line search, so the Wolfe conditions see a stationary target, then refreshed against the updated model between optimiser blocks. Within a shell, σ_A (equivalently $v = 1 - \sigma_A^2$) is chosen to minimise the Rice / folded-normal negative log-likelihood of Eq. S26 at mean $\alpha|F_c|$, evaluated on a bounded geometric grid with nested zoom (17 candidates over three stages). The grid carries no data-dependent branching and locates the same optimum as the one-dimensional root find used in `mmtbx`. The per-shell values are then stabilised by a single DerSimonian–Laird shrinkage toward a smooth curve $-\ln \sigma_A = a + b\, d^{*2}$ (with $b \geq 0$) fitted across all shells, which damps noisy, sparsely populated shells. Two backstops guard degenerate shells: σ_A is capped at 0.99 (a floor on the model-error variance), and a lower bound on σ_A is chosen shell by shell so that the derived $\alpha \geq 0.1$. This lower bound is a numerical backstop, not a statement that the formal Wilson limit occurs at $\alpha = 0.1$.

Per reflection, the four shell curves σ_A, $\log \Sigma_N$, $\log \Sigma_P$ and S^2 are interpolated in d^{*2} and α and β are *re-derived* from them ($\beta_{\text{model}} = (1 - \sigma_A^2)\Sigma_N$, $\beta = \beta_{\text{model}} + S^2$, $\alpha = \sigma_A \sqrt{\Sigma_N / \Sigma_P}$). Interpolating β directly is avoided: it can yield a value consistent with no $\sigma_A \leq 1$, introducing an inconsistency in the decomposition. A final per-reflection $\alpha \geq 0.1$ clamp remains as a degenerate-shell numerical backstop.

5.3 Relationship between the targets, and choice of default

The four model-error modes (`ml`, `ml_noalpha`, `ml_full`, and `nll_beta`) use one σ_A estimator through three likelihood shapes. Setting $\alpha = 1$ reduces `ml` to `ml_noalpha`; taking the large-signal ($2F_oF_c/\Sigma \gg 1$) Gaussian limit of `ml_noalpha` gives `nll_beta`, whose amplitude variance is $\varepsilon\beta$ folded to the amplitude convention ($\Sigma/2$ acentric, Σ centric); replacing that σ_A-derived variance by the measurement variance σ^2 and dropping the modified-Bessel structure of Eq. S26 gives the separate `nll` mode, which needs no σ_A estimate and behaves like least squares. `ml_full` is `ml` with the measurement error marginalised rather than folded into Σ. Within the four model-error modes,

the estimator and shrinkage procedure are shared, so their comparison isolates the stated likelihood choices.

We keep the D-coupled `ml` as the default because it is the appropriate conditional distribution, with an empirical per-shell error model that accounts for model incompleteness and phase uncertainty as established maximum-likelihood programs do. On TorchRef's validated benchmark set ($n = 750$), the median R_{free} is 0.3170 for `ml`, compared with 0.3190 for `ml_noalpha`, 0.3184 for `ml_full`, and 0.3203 for `nll_beta`. The paired median change from `ml_noalpha` to the coupled `ml` target is -0.0012, so these data support, rather than penalise, the coupled default.

5.4 Difference refinement for time-resolved data

Difference refinement, in which the residual between observed and calculated structure factor amplitude *differences* between two data sets is minimised, was introduced by Berendzen & Terwilliger (Berendzen & Terwilliger, 1995) for pairs of nearly isomorphous, fully-occupied structures (native vs variant, native vs mutant); their companion Bayesian formulation (Terwilliger & Berendzen, 1996) extends the residual to handle partial correlation of model errors, interpolating smoothly between difference refinement and independent refinement of the two models. TorchRef adopts the underlying data residual of (Berendzen & Terwilliger, 1995) (the squared difference of amplitude differences, inverse-variance weighted by the propagated experimental uncertainty) and extends the framework to time-resolved data where the light state is only *partially* populated. Three elements are not in the original Berendzen & Terwilliger framework and are introduced here:

1. a *mixed-model parameterisation* of the light timepoint as a population-weighted combination of dark and light structures, so the refinement target depends on the experimental light-state fraction rather than assuming full occupancy;
2. a *spike-and-slab prior* on coordinate change (Mitchell & Beauchamp, 1988) that decides per atom whether a displacement is real or noise-driven, replacing the smooth Gaussian shrinkage implicit in Terwilliger's correlation-interpolated residual;
3. a *warmup protocol* that defers the prior to a subsequent cleanup pass when large displacements are expected.

Mixed-model parameterisation. The crystal at the photo-excited timepoint is modelled as a population mixture of dark and light structures with light fraction w, as in Eq. S31:

$$F_{\text{calc}}^{\text{mix}}(\mathbf{h}) \;=\; (1-w)\,F_{\text{calc}}^{\text{dark}}(\mathbf{h}) + w\,F_{\text{calc}}^{\text{light}}(\mathbf{h}). \tag{S31}$$

The dark coordinates and B-values are frozen and act as a fixed reference; only the light-state parameters are refined. w defaults to the experimental light-state occupancy estimate (e.g. 0.22 for the pdb_00007yyz 10 μs timepoint) and is frozen by default; CLI flags expose fraction refinement when desired. Refining w from a single timepoint is not recommended: the occupancy is only weakly

determined, so a refined value must be interpreted with care. The same mixed-model primitive generalises to many timepoints through the `ModelCollection` infrastructure (Section 9).

An explicit single-timepoint sensitivity study supports this default. When w was refined from starting values 0.10, 0.22, and 0.40, all three runs converged within one macrocycle to $w = 0.083 \pm 0.001$, rather than recovering the independent estimate. The resulting chromophore (IBL) real-space difference-map correlation was 0.805, compared with 0.845 when w was fixed at 0.22. In a frozen-w scan from 0.05 to 0.55, the local map correlation had a broad maximum around 0.22 (scan value 0.852), while the global R_{free} profile was shallow and reached its minimum near 0.10. Because the σ_A term was itself estimated from free reflections, the R_{free} curve is read only as a profile shape. The independent occupancy is therefore retained rather than treated as a measurement.

Difference data likelihood. The data likelihood is a Gaussian negative log-likelihood (NLL) on amplitude differences with inverse-variance weighting from the propagated experimental uncertainty. TorchRef implements this in a generalised, mean-referenced form that applies to an arbitrary collection of N related datasets: each dataset's amplitude is compared against the mean amplitude over the collection, so that no single dataset is privileged as the reference. For dataset i, with observed amplitudes $F_o^{(i)}$ (uncertainty $\sigma_o^{(i)}$) and model amplitudes $|F_{\text{calc}}^{(i)}|$, as in Eq. S36:

$$\Delta F_o^{(i)} = F_o^{(i)} - \tfrac{1}{N} \sum_j F_o^{(j)} \tag{S32}$$

$$\Delta F_c^{(i)} = \left|F_{\text{calc}}^{(i)}\right| - \tfrac{1}{N} \sum_j \left|F_{\text{calc}}^{(j)}\right| \tag{S33}$$

$$\sigma_i^2 = \sigma_o^{2,(i)} \left(1 - \tfrac{2}{N}\right) + \tfrac{1}{N^2} \sum_j \sigma_o^{2,(j)} \tag{S34}$$

$$\sigma_{\text{safe}}^{2,(i)} = \max\!\left(\sigma_i^2,\ (0.1 \cdot \text{median}\, \sigma)^2\right) \tag{S35}$$

$$\mathcal{L}_{\text{diff}} = \sum_i \sum_{\mathbf{h}} \left[\frac{1}{2} \frac{(\Delta F_o^{(i)} - \Delta F_c^{(i)})^2}{\sigma_{\text{safe}}^{2,(i)}} + \log \sigma_{\text{safe}}^{(i)} + \tfrac{1}{2} \log(2\pi) \right] \tag{S36}$$

where σ_i^2 is the variance of $\Delta F_o^{(i)}$ under independent per-dataset errors and the median in the σ_{safe} floor is taken over the aligned reflection set. For the two-state case $N = 2$ (dark/light), the two centred rows together give the familiar light-minus-dark amplitude-difference residual with propagated variance $\sigma_o^{2,\text{light}} + \sigma_o^{2,\text{dark}}$, but with an overall factor of two in the summed NLL. It thus has the same minimiser, with loss and gradient magnitudes twice those of the single-difference convention. The time-resolved showcase below is this $N = 2$ case. No phases enter the refinement target itself: dark-state phases are used downstream to produce the difference electron density map for visualisation, but the target compares scalar amplitudes only.

Spike-and-slab prior on coordinate change. Terwilliger's framework uses a Gaussian prior on the dark-to-light coordinate change $\Delta\mathbf{r}$, which damps real conformational changes quadratically

along with noise. TorchRef instead uses a *spike-and-slab* prior (Mitchell & Beauchamp, 1988), a Bayesian model-averaging prior that mixes a "static" hypothesis (the atom did not move) with a "dynamic" hypothesis (the atom moved by an unconstrained amount). Marginalising the static-vs-dynamic indicator, the per-atom log-prior is given by Eq. S37:

$$\mathcal{L}_{\text{sim}}(\Delta\mathbf{r}) \;=\; \sum_{\text{atoms}} -\log\Big(\exp\big(-\tfrac{\|\Delta\mathbf{r}\|^2}{2\sigma^2}+\alpha\big)+1\Big), \tag{S37}$$

applied to each atom's 3-D Cartesian displacement $\Delta\mathbf{r} = \mathbf{r}^{\text{light}} - \mathbf{r}^{\text{dark}}$. $\sigma^2 = B_{\text{dark}}/(8\pi^2)$ is the per-axis variance of the isotropic 3-D Gaussian implied by the dark-state isotropic B-value (detached from the autograd graph); the corresponding mean-square 3-D displacement is $\mathbb{E}[\|\Delta\mathbf{r}\|^2] = 3\sigma^2$. α is a global log prior-odds hyperparameter (default $\alpha = 2$). Behaviour (per atom):

- For $\|\Delta\mathbf{r}\| \ll \sigma$, the prior is quadratic in $\|\Delta\mathbf{r}\|$ and behaves like a Gaussian, so small noise-driven displacements are suppressed.
- For $\|\Delta\mathbf{r}\| \gg \sigma$, the prior plateaus to a constant (gradient $\to 0$). This is the key qualitative difference from Terwilliger's prior: real coordinate changes that exceed the per-atom dark-state width are not damped, so the refinement can recover large rearrangements without the conservative "shrinkage towards dark" that pure-Gaussian priors impose.
- The crossover is at $\|\Delta\mathbf{r}\| \approx \sigma\sqrt{2\alpha}$ ($\approx 2.0\,\sigma$ for $\alpha = 2$).

Relation to top-out reference-model restraints. The negative log-prior of Eq. S37 is a top-out (redescending) potential (quadratic in $\|\Delta\mathbf{r}\|$ near zero, plateauing to a constant with vanishing gradient beyond the crossover), and in that respect belongs to the same family as the Geman–McClure restraint used for external / reference-model restraints in REFMAC5 via ProSMART (Nicholls *et al.*, 2014) and the analogous top-out restraints in `phenix.refine` and ISOLDE, all of which restrain small deviations while declining to penalise large, genuine ones. The specific forms differ: REFMAC's Geman–McClure is a rational top-out, ISOLDE's adaptive restraint is a tunable generalisation of common robust losses (Croll & Read, 2021), and Eq. S37 is a log-sum-exp (softplus) form, so it is a distinct member of the family rather than a re-derivation of an existing one. It also differs in context and in form. *Context:* those are reference-model restraints that regularise geometry toward a fixed external structure, whereas the same functional shape serves here as a prior on the dark→light coordinate *change*, a change-detection prior rather than a geometry regulariser. *Form:* (i) it is derived as the marginal of an explicit two-component Bayesian mixture, so its crossover follows from interpretable quantities (the log prior-odds α and a physically motivated spike width) rather than a free tuning constant; (ii) the spike width $\sigma^2 = B_{\text{dark}}/(8\pi^2)$ is set *per atom* from the dark-state B-value, so the noise threshold adapts to each atom's own thermal displacement, whereas a Geman–McClure scale is typically a single global constant; (iii) because it is a mixture, one can read off a per-atom posterior probability of genuine movement, a natural output for difference / time-resolved refinement; and (iv) the quadratic spike shrinks

sub-threshold displacements toward zero more sharply than a smooth Geman–McClure, giving a cleaner noise-vs-signal separation.

Warmup considerations and the α/weight schedule. At $\|\Delta\mathbf{r}\| \ll \sigma$ the spike-and-slab gradient pulls the light model back towards dark; if the prior is active from cycle 0 (when the light model still *is* the dark model) it can overpower the X-ray difference signal and prevent the model from displacing at all. The default similarity weight (1.0) is chosen as a gentle prior that biases towards parsimony without dominating the X-ray signal in typical small-to-moderate cases. For workflows that anticipate larger displacements (e.g. ligand translation of several Å between dark and light states), the recommended protocol is to run an initial pass with `--similarity-weight 0` (X-ray difference only, unconstrained), then re-enable the prior at 1.0 or 3.0 for a cleanup pass.

Independently, the difference-target weight follows an annealing schedule (default `--weight-schedule 5,3,2`) that starts high to escape the dark-state local minimum and relaxes once the light model approaches the light-state basin.

Showcase protocol (pdb_00007yyz). The published azo-CA4/tubulin difference-refined model (Wranik *et al.*, 2023) was produced by an iterative protocol rather than a single automated run: an initial automated difference-refinement pass with `torchref.difference-refine`; two cycles of alternating manual model building in Coot using the phase-corrected, extrapolated structure factors written out by the tool; and a final automated refinement pass with both the difference-target weight and the similarity weight set to 1 to equilibrate into a favourable geometry.

6 Geometric restraints

6.1 Restraint construction

Geometric restraints are built once at setup and cached for the model's lifetime. For each restraint type (bond, angle, torsion, plane, chiral), each residue's atom names are matched against the relevant rows of the CIF restraint dictionary in a single pass over the residue list, and the resulting restraint indices are sorted by lead atom index for cache-friendly access during forward evaluation. Inter-residue links (peptide bonds, disulfides, and PDB `LINK` records) are handled in a separate pass over consecutive residue pairs.

Monomer library. Ideal geometry parameters are read from the CCP4 Monomer Library (Long *et al.*, 2017), pinned to a specific commit hash for reproducibility and resolved, in order, from an environment override, bundled package data, a user cache, or an on-demand download from the pinned commit. Non-standard residues are downloaded once and reused.

Disulfide cross-links. Disulfide bonds are detected by a pairwise SG–SG distance scan with a 2.5 Å cutoff (excluding intra-residue pairs) and routed through dedicated angle and torsion builders

that use the `disulf` link definition from the monomer library. Disulfide torsions are emitted with period = 2 and are folded into the torsion target's unimodal von Mises NLL branch alongside other torsion restraints. PDB `LINK` records share the same code path with explicit de-duplication against auto-detected disulfides.

6.2 Restraint functional forms

All geometric restraints are implemented as Gaussian negative log-likelihoods of the form of Eq. S38,

$$\mathrm{NLL} = \frac{1}{2}\left(\frac{x - x_0}{\sigma}\right)^2 + \log\sigma + \frac{1}{2}\log(2\pi), \tag{S38}$$

with ideal values x_0 and standard deviations σ taken from the CCP4 monomer library (and therefore varying by bond/angle/atom type).

Bond length. $x = \|\mathbf{r}_j - \mathbf{r}_i\|$, x_0 the ideal bond length.

Bond angle. With $\mathbf{v}_1 = \mathbf{r}_a - \mathbf{r}_b$, $\mathbf{v}_2 = \mathbf{r}_c - \mathbf{r}_b$, the angle is given by Eq. S39:

$$\theta = \arccos\left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\|\,\|\mathbf{v}_2\|}\right), \tag{S39}$$

with the cosine clamped so that the gradient stays finite at collinearity.

Torsion (unimodal, von Mises). With concentration $\kappa = \mathrm{clamp}(1/\sigma^2, 10^{-3}, 10^4)$ and the deviation $\Delta\omega$ from the ideal angle wrapped to its smallest-magnitude n-fold-equivalent representative (period n up to 6; $n = 2$ for disulfides), the NLL takes the form of Eq. S40:

$$\mathrm{NLL} = \log(2\pi) + \log I_0(\kappa) - \kappa\cos(\Delta\omega), \tag{S40}$$

where $\log I_0(\kappa) = \mathrm{logi0e}(\kappa) + \kappa$ is evaluated in a numerically stable form at large κ.

Peptide ω (cis/trans mixture). A two-component von Mises mixture as in Eq. S42,

$$P(\omega) = w_{\mathrm{trans}}\,\mathrm{VM}(\omega; \pi, \kappa) + w_{\mathrm{cis}}\,\mathrm{VM}(\omega; 0, \kappa), \tag{S41}$$

$$\mathrm{NLL} = \log(2\pi) + (\mathrm{logi0e}(\kappa) + \kappa) - \mathrm{logsumexp}\big(\log w_{\mathrm{trans}} - \kappa\cos\omega,\ \log w_{\mathrm{cis}} + \kappa\cos\omega\big), \tag{S42}$$

with $w_{\mathrm{cis}} = 0.05$ for pre-proline omegas (the C(i)–N(i+1) bond when residue i+1 is proline; ∼5% cis in the PDB) and 0.0005 otherwise (∼0.05% cis), with $w_{\mathrm{trans}} = 1 - w_{\mathrm{cis}}$.

Planarity. The plane normal $\hat{\mathbf{n}}$ is the eigenvector of smallest eigenvalue of the 3×3 covariance of the centred coordinates (equivalently the right singular vector of smallest singular value), detached from the graph; the per-atom deviation is $\mathrm{dev}_a = \hat{\mathbf{n}} \cdot (\mathbf{x}_a - \bar{\mathbf{x}})$, restrained to zero by a Gaussian NLL (one Gaussian per member atom of each plane, with per-atom σ from the monomer library, summed across atoms of the plane).

Chiral volume. With $\mathbf{v}_i = \mathbf{x}_i - \mathbf{x}_{\text{center}}$, the signed chiral volume of Eq. S43,

$$V = \mathbf{v}_1 \cdot (\mathbf{v}_2 \times \mathbf{v}_3), \tag{S43}$$

restrained to the ideal signed volume by a Gaussian NLL; achiral centres (which carry no signed ideal, stored as 0) instead restrain the magnitude $|V|$ towards 2.5 Å^3.

Ramachandran. A precomputed NLL $= -\log P(\varphi, \psi \mid \text{type})$ surface on a 360×360 grid at 1° resolution covering $[-180^\circ, +180^\circ)$ (six residue-type surfaces) is evaluated by bilinear interpolation, differentiable in the interpolation fraction.

6.3 Hydrogen handling for non-bonded repulsion

The non-bonded repulsion target receives a hydrogen-aware extension when the model includes (or is expanded with) explicit hydrogens. Hydrogen positions are recomputed from heavy-atom positions on every forward pass, using the same riding-H principle as the AMBER target (Section 8.2). Each hydrogen is classified into one of several geometric placement types, with the placement parameters stored as non-trainable buffers.

A hydrogen-specific non-bonded term (`NonBondedHTarget`) is added to the loss when hydrogens are present. The exclusion set is hydrogen aware: for each H, the 1–2 exclusion is H–parent and the 1–3 exclusion set includes H–{heavy neighbours of parent}. Exclusions are applied at pair-construction time. A uniform van der Waals radius of 1.20 Å is used for every riding H.

Hydrogen contribution to structure factors. By default, the `Model` class strips hydrogens at load (`strip_H = True`); riding hydrogens used by the van der Waals (VDW) target are *not* added to the structure factor sum. An opt-in path (`Model.exclude_H_from_sf` together with `Model.hydrogenate()`) lets explicit, pre-placed hydrogens from the monomer library participate in F_{calc}; those hydrogens are stored in the PDB DataFrame and are unrelated to the riding-H topology used by VDW or AMBER.

7 ADP restraints

The default composite ADP regulariser is the sum of a bonded-atom similarity term, a k-nearest-neighbour (kNN) locality term, and a population-level shifted inverse-gamma term; a rigid-bond term is available but not on by default.

Bonded-atom similarity. The B-values of covalently bonded atoms are restrained to be similar via the Gaussian NLL of Eq. S44. This term acts on bonded pairs; through-space smoothness is

handled by the kNN locality term below:

$$\mathrm{NLL}_{\mathrm{SIMU}} = \sum_{\text{pairs}} \left[\frac{1}{2} \left(\frac{B_i - B_j}{\sigma_{\mathrm{SIMU}}} \right)^2 + \log \sigma_{\mathrm{SIMU}} + \frac{1}{2} \log(2\pi) \right], \tag{S44}$$

with default $\sigma_{\mathrm{SIMU}} = 2.0$ Å^2. For models containing anisotropic atoms the restraint operates on the unified per-atom $\mathbf{U}_6$ tensor and splits into two channels. A *magnitude* channel restrains the equivalent isotropic $B_{\mathrm{eq}} = \frac{8\pi^2}{3} \mathrm{tr}(\mathbf{U})$ with the same Gaussian NLL as Eq. S44, so it reduces exactly to the isotropic form when every atom is isotropic. A *deviatoric* channel restrains the traceless part of the **B**-tensor $\mathbf{D}_i = 8\pi^2 \left(\mathbf{U}_i - \frac{1}{3}\mathrm{tr}(\mathbf{U}_i)\mathbf{I} \right)$ component-wise through a quadratic penalty (the constants $\log \sigma + \frac{1}{2} \log 2\pi$ are dropped for the fixed-σ deviatoric channel) with sigma $\sigma_{\mathrm{SIMU,\,aniso}}$ (default 1.0 Å^2); explicitly, $\mathrm{NLL}_{\mathrm{dev}} = \frac{1}{2} \sum_{\text{pairs}} \sum_a w_{\mathrm{comp},a} \left((D_{i,a} - D_{j,a}) / \sigma_{\mathrm{SIMU,\,aniso}} \right)^2$ with weights $w_{\mathrm{comp}} = (1, 1, 1, 2, 2, 2)$ on the six unique components (equivalent to the Frobenius norm of the symmetric 3×3 **D**-tensor, with off-diagonal entries counted twice). Because $\mathbf{D}_i \equiv 0$ for an isotropic atom, iso–aniso pairs are handled natively with no special casing.

kNN locality (spatial smoothness on $\log B$). Over each atom's $k = 50$ nearest neighbours, with inverse-distance weights $w_{ij} = 1/(d_{ij} + 10^{-6})$, the locality term reads as Eq. S45:

$$L_{\mathrm{loc}} = \sum_i \sum_{j \in \mathrm{kNN}(i)} w_{ij} \left(\frac{\log B_i - \log B_j}{0.5} \right)^2. \tag{S45}$$

The anisotropic extension keeps the same form on the magnitude channel (replacing B with B_{eq}) and adds an anisotropy channel on the *fractional* deviatoric tensor $\mathbf{D}_i / B_{\mathrm{eq,i}}$, i.e. the deviatoric part normalised by the equivalent isotropic B, so that the new channel lives on the same dimensionless scale as $\log B_{\mathrm{eq}}$:

$$L_{\mathrm{loc}}^{\mathrm{aniso}} = \sum_i \sum_{j \in \mathrm{kNN}(i)} w_{ij} \left[\left(\frac{\log B_{\mathrm{eq},i} - \log B_{\mathrm{eq},j}}{0.5} \right)^2 + \left\| \frac{\mathbf{D}_i / B_{\mathrm{eq},i} - \mathbf{D}_j / B_{\mathrm{eq},j}}{\sigma_{\mathrm{aniso}}} \right\|_F^2 \right], \tag{S46}$$

with $\sigma_{\mathrm{aniso}} = 0.5$ by default, comparable to the fixed log-sigma of the magnitude channel, and B_{eq} clamped at 10^{-3} Å^2 for numerical safety inside both the log and the fractional-anisotropy ratio. As with the bonded-atom restraint, the magnitude channel reduces exactly to Eq. S45 when every atom is isotropic.

Population-level (shifted inverse-gamma). This is a prior on the *distribution* of B-values across the whole model, not on individual atoms. Following the theoretical and empirical analysis of Masmaliyeva & Murshudov (Masmaliyeva & Murshudov, 2019; Masmaliyeva *et al.*, 2020), which shows the macromolecular B-value distribution to be well described by a shifted inverse-gamma distribution (and better so than by a log-normal), we restrain the B-values toward that family. Writing $x_i = B_i - B_0$ for the shifted B-value of atom i, the per-atom negative log-likelihood under an inverse-gamma(α, β) prior, offset to vanish at its mode $x_{\mathrm{mode}} = \beta/(\alpha + 1)$, gives the restraint of

Eq. S47:

$$L_{\mathrm{ADP}}^{\mathrm{dist}} = \sum_i (\alpha+1)\,\log\frac{x_i}{x_{\mathrm{mode}}} + \beta\left(\frac{1}{x_i} - \frac{1}{x_{\mathrm{mode}}}\right), \qquad x_i = B_i - B_0. \tag{S47}$$

The shift B_0 is fixed at 0 by default (a non-zero shift is degenerate with overall sharpening/blurring), and the shape $\alpha = 3.5$ is fixed, consistent with the PDB-wide B-value distributions of Masmaliyeva & Murshudov; it sets the log-width of the distribution, $\mathrm{std}(\log B) = \sqrt{\psi_1(\alpha)} \approx 0.575$ (with ψ_1 the trigamma function), independent of scale. The scale β is re-estimated on every forward pass by moment matching, $\beta = (\alpha-1)\langle x\rangle$, but is *detached* from the autograd graph, so the restraint gradient acts only on the *spread* of the B-value distribution and cannot drive the overall B level: it is **not** a Wilson-B or absolute-scale prior. It carries the default ADP-group weight (0.02) and is on by default. This shifted inverse-gamma form replaces an earlier log-normal (Gaussian-in-log B) width term; besides being the better-justified target, it stays finite for a uniform B distribution (where the log-normal KL diverged) and is monotone in $\mathrm{std}(\log B)$, so it cannot reward spreading the distribution.

Rigid-bond (DELU; optional). For a bonded pair, isotropic $\Delta z = (B_i - B_j)/(8\pi^2)$; anisotropic $\Delta z = \hat{\mathbf{l}}^{\mathsf{T}}\mathbf{U}_i\hat{\mathbf{l}} - \hat{\mathbf{l}}^{\mathsf{T}}\mathbf{U}_j\hat{\mathbf{l}}$ with $\hat{\mathbf{l}} = (\mathbf{r}_j - \mathbf{r}_i)/\|\mathbf{r}_j - \mathbf{r}_i\|$; restrained by a Gaussian NLL on Δz with default $\sigma_{\mathrm{DELU}} = 0.004$ Å^2.

8 AMBER14/GAFF2 force-field target

TorchRef provides a differentiable AMBER14/GAFF2 force-field (Maier *et al.*, 2015; Wang *et al.*, 2004) energy as an experimental opt-in alternative to the empirical geometric restraints described above. The implementation is a proof of concept; it inherits the limitations of an MD-style energy applied to a single static conformation and is not used in the AlphaFold benchmark of the main text.

8.1 Bridge between OpenMM and PyTorch autograd

The AMBER14/GAFF2 potential energy and its analytical forces are evaluated by OpenMM (Eastman *et al.*, 2017). A custom `torch.autograd.Function` wraps the OpenMM call so that the energy is differentiable with respect to atomic Cartesian coordinates via the force–gradient identity of Eq. S48,

$$\frac{\partial E_{\mathrm{AMBER}}}{\partial \mathbf{x}} \;=\; -\,\mathbf{F}_{\mathrm{OpenMM}}(\mathbf{x}), \tag{S48}$$

with $\mathbf{F}_{\mathrm{OpenMM}}$ the analytical force vector returned by OpenMM's `getState(getForces=True)`. No finite-difference gradient is used. A per-atom force clamp is applied to the forces returned to autograd, preventing near-divergent Lennard-Jones forces (e.g. from symmetry-related clashes) from destabilising the first optimiser steps; the resulting minor inconsistency between the unclamped forward energy and the clamped backward gradient resolves after the first macrocycle.

The OpenMM `CMMotionRemover` is removed from the system so that per-atom forces are exposed

without spurious centre-of-mass coupling, and the platform is selected to match the model's device (CUDA, OpenCL, or CPU fallback).

8.2 Riding-hydrogen model

Hydrogen positions are not refined as independent parameters. For each hydrogen H, an orthonormal local frame is built from its parent heavy atom $\mathbf{p}$ and two heavy-atom neighbours $\mathbf{n}_1$, $\mathbf{n}_2$ as in Eq. S49,

$$\mathbf{e}_1 = \widehat{\mathbf{n}_1 - \mathbf{p}}, \quad \mathbf{e}_2 = \widehat{(\mathbf{n}_2 - \mathbf{p})_{\perp \mathbf{e}_1}}, \quad \mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2, \quad \mathbf{r}_H = \mathbf{p} + l_x \mathbf{e}_1 + l_y \mathbf{e}_2 + l_z \mathbf{e}_3, \tag{S49}$$

where (l_x, l_y, l_z) are captured once at construction from the input geometry and stored as frozen buffers. Hydrogen positions are *recomputed every forward pass* through this map, so the analytical AMBER force on each H is automatically redistributed onto its parent and the two frame-reference atoms via the exact local-frame Jacobian (handled by autograd; no finite-step approximation). When the parent has only one heavy neighbour, a second reference atom is selected one bond further out (a heavy neighbour of $\mathbf{n}_1$); only when this also fails do hydrogens fall back to a rigid translation $\mathbf{r}_H = \mathbf{p} + \mathbf{o}$ where $\mathbf{o}$ is the construction-time offset.

The riding-hydrogen model is the same physical idea as the riding-H treatment in SHELX (Sheldrick, 2015) and matches what the VDW non-bonded target uses (Section 6.3), but the AMBER implementation maintains its own riding scheme tied to the OpenMM atom indexing. Model hydrogens contribute to the energy only via the riding placement, not as independent OpenMM atoms.

8.3 Non-standard residues

Heteroatoms not in the AMBER14 standard residue set are parameterised automatically: `antechamber` (Wang *et al.*, 2006) assigns GAFF2 atom and bond types and partial charges, and `parmchk2` generates missing force-field parameters. Results are cached on disk, keyed on the residue topology, so a given non-standard residue is parameterised at most once across runs. Heavy-atom-only ligands are auto-protonated against the monomer library before `antechamber` so that GAFF2 typing and `sqm` see satisfied valences.

Altloc atoms are filtered before building the OpenMM system: only the primary conformation (altloc `''` or `A`) enters, and the target raises explicitly if the model still contains residual altlocs.

8.4 Loss and normalisation

The reported loss is the total energy divided by the number of model atoms when `normalize_by_atoms = True` (default), giving units of $\mathrm{kJ\,mol^{-1}\,atom^{-1}}$, so the target is roughly intensive and can be combined with the per-reflection X-ray and geometry NLLs through the standard weighting hierarchy. When OpenMM atoms have no model counterpart (e.g. heavy atoms of a non-standard residue

that the input model does not contain), the missing positions are filled from a construction-time snapshot rather than zero, so the energy does not blow up.

9 Multi-dataset and multi-model infrastructure

Many refinement workflows pair a shared set of structural parameters with multiple datasets: time-resolved series, multi-state ensembles, condition series (temperature, pH, ligand-on/ligand-off), MAD/SAD experiments, and multi-crystal merging all instantiate the same abstract problem. TorchRef provides paired data and model collection objects for this regime, with a single shared set of scaling parameters across all data–model pairs. This is the infrastructure underlying the difference refinement workflow of Section 5.4 and the ensemble extensions in the `experimental` module. Throughout, *dataset* denotes one member of a collection; time-resolved *timepoints* are the most prominent specialisation.

9.1 MixedModel

A `MixedModel` combines N `ModelFT` instances with a population fraction vector obtained by softmax normalisation, Eq. S50,

$$F_{\text{calc}}^{\text{mix}}(\mathbf{h}) \;=\; \sum_{i=1}^{N} w_i \, F_{\text{calc}}^{(i)}(\mathbf{h}), \qquad w_i \;=\; \frac{\exp(p_i)}{\sum_j \exp(p_j)}, \tag{S50}$$

where the latent p_i are refinable and $\sum_i w_i = 1$ by construction. Each component model carries its own coordinates, ADPs, occupancies, and solvent contribution; fractions can be frozen at experimental estimates or refined. This is the primitive used for any partial-occupancy or multi-state refinement.

9.2 ModelCollection

A `ModelCollection` is a named dictionary of mixed models, one per dataset in the collection. Critically, the *base structural models are shared across datasets*, so refining N datasets does not duplicate the structural parameters N times; only the per-dataset fractions are independent. External fraction sources (for example, a kinetic occupancy model parameterised by rate constants in the time-resolved case, or condition-dependent occupancies elsewhere) can override the softmax-derived fractions while preserving gradient flow back through the upstream model.

9.3 DatasetCollection

A `DatasetCollection` aligns multiple `ReflectionData` objects to a common HKL set. Alignment uses the first/reference dataset as the canonical HKL list; subsequent datasets are remapped to this list, and reflections missing in any dataset are masked out per dataset (rather than dropped

globally). The collection keys are matched against the `ModelCollection` keys so that targets pair each dataset with its corresponding mixed model automatically.

9.4 CollectionScaler and shared solvent

A `CollectionScaler` (Section 4.7) carries a single shared set of scale parameters ($\mathbf{c}_{\text{iso}}$, $\mathbf{U}$, k_{sol}, s_h^2, n, φ_{offset}) across all data–model pairs in a collection. Per-component solvent models are created for each base structural model in the `ModelCollection`; for any dataset in the collection the solvent contribution is the linear combination of these per-component $F_{\text{sol}}^{(i)}$ weighted by the same population fractions as the structural models, as in Eq. S51:

$$F_{\text{sol}}^{\text{mix}}(\mathbf{h}) \;=\; \sum_i w_i \, F_{\text{sol}}^{(i)}(\mathbf{h}). \tag{S51}$$

A primary solvent model owns the learnable solvent parameters; the non-primary models inherit them through frozen references. Sharing the scale across data–model pairs prevents artificial per-dataset scale drift that would otherwise contaminate the difference signal in time-resolved refinement.

9.5 Collection targets

Three multi-dataset X-ray targets operate on paired `DatasetCollection` + `ModelCollection` instances:

- `CollectionDifferenceTarget`: a mean-based difference target that compares each dataset's amplitudes against the cross-dataset mean, with proper covariance-corrected variance $\text{Var}(F_i - \overline{F}) = \sigma_i^2(1 - 2/N) + \sum_j \sigma_j^2/N^2$.
- `CollectionRiceTarget`: a multi-dataset Rice maximum-likelihood amplitude target with $\beta = \sigma^2$ (per-reflection).
- `CollectionMLTarget`: a σ_A-style ML target with a *single shared per-reflection* β (Luzzati α fixed at 1) estimated by ML on *pooled* free reflections across all datasets. This produces a consistent error model across the collection, which is necessary for headline statistics to be comparable.

All three targets are vectorised on stacked (N, n_{hkl}) tensors and reuse the same per-dataset scaling output.

10 Loss aggregation and weighting

Targets are registered under a two-level group/component hierarchy (`xray`, `geometry`, `adp`, plus per-target components such as `geometry/bond` or `xray/difference`). At evaluation, each target's loss is multiplied by the product of its group weight and component weight, and zero-weight targets

are skipped entirely (no forward pass, no cache). If any target returns a non-finite value the aggregate loss returns $+\infty$, which causes the strong Wolfe line search to backtrack to the last finite point.

The default weighting is static and is stored in `DEFAULT_GROUP_WEIGHTS` as in Eq. S52:

$$w_{\text{xray}} = 1.0, \qquad w_{\text{geometry}} = 0.2, \qquad w_{\text{adp}} = 0.02. \tag{S52}$$

The geometry NLLs (CCP4 Monomer Library Gaussian targets (Long *et al.*, 2017); protein bond/angle targets derive from Engh–Huber (Engh & Huber, 1991)) enter at $0.2\times$ the X-ray loss, and the ADP prior at $0.02\times$. These defaults were calibrated on a stratified subset of the AlphaFold benchmark (see the main-text Validation section and Fig. S1). In the completed benchmark, coordinate-geometry RMSZ is near REFMAC5, whereas the main-chain bonded-B RMSZ remains higher than in the reference engines and the absolute ADP scale requires separate follow-up.

Weighting is delivered through a `ManualWeighting` scheme that returns a fixed weights dictionary independent of refinement state. `ManualWeighting` accepts both group-level keys (`"geometry"`) and per-component overrides (`"geometry/bond"`); the defaults populate only the group level. The earlier adaptive schemes (policy-based weighting, Evolution Strategies, random-perturbation) have been deprecated in favour of the simpler static scheme.

11 Optimisation and refinement cycle

11.1 Optimiser

The default optimiser is L-BFGS (torch.optim.LBFGS, learning rate 1.0, ≤ 20 inner iterations, history size 100, `line_search_fn="strong_wolfe"`). The curvature history is cleared between phases, because the active parameter set and the loss landscape change. There is no additional global convergence test across macrocycles. Within each block, PyTorch L-BFGS stops when its gradient or parameter/loss-change tolerance is met, when the inner iteration/evaluation budget is exhausted, or when the strong-Wolfe line search cannot accept a further step. The macrocycle count is user-set. TorchRef remains compatible with any PyTorch optimiser; alternative optimisers for escaping local minima (Lanczos-Hessian exploration, BAOAB Langevin with annealing, Metropolis simulated annealing) are also provided.

11.2 Macrocycle structure

Each macrocycle starts with a scaler refinement, re-absorbing any overall B-trend from the previous cycle's coordinate and ADP updates before atomic parameters are refined again (mitigating the known ADP–overall-B degeneracy). Concretely, each macrocycle performs, in order:

1. **Scaler refinement** (refine_scaler), including a bulk-solvent update. It fits scale parameters only and does not flag or mask reflections.

2. **Coordinate refinement** (refine_xyz).

3. **ADP/occupancy refinement** (refine_adp): isotropic B, anisotropic $\mathbf{U}$, and occupancies.

After the specified number of macrocycles, a final scaler refinement is performed before output. Per-cycle statistics (R_{work}, R_{free}, the R-gap, and geometry/ADP metrics) are logged.

11.3 Multi-resolution rigid-body refinement

Rigid-body refinement uses a coarse-to-fine resolution schedule with a target-function switch. Per-chain rigid-body parameters (rotation and translation) replace the free Cartesian coordinates, as in Eq. S53,

$$\mathbf{x}_{i,c} \;=\; \mathbf{R}(\boldsymbol{\alpha}_c)\,(\mathbf{x}^0_{i,c} - \boldsymbol{\mu}_c) + \boldsymbol{\mu}_c + \mathbf{t}_c, \tag{S53}$$

with $\boldsymbol{\alpha}_c = (\alpha_c, \beta_c, \gamma_c)$ the Euler angles of chain c, $\mathbf{t}_c$ its translation, $\boldsymbol{\mu}_c$ a mass-weighted centroid, and $\mathbf{x}^0_{i,c}$ the reference coordinates at the start of the current LBFGS sub-cycle. The convention is $R_z(\gamma)R_y(\beta)R_x(\alpha)$ (extrinsic XYZ Euler $\equiv$ intrinsic ZYX), deliberately chosen to avoid the gimbal-lock singularity at the origin that ZYZ Euler would impose after each parameter reset.

The resolution schedule defaults to a coarse-to-fine sequence of cutoffs derived from the native $d_{\min}$. At each cutoff:

- For $d_{\min} > 6$ Å, the X-ray target is the `mmtbx`-style `ls_wunit_k1` (Afonine *et al.*, 2012): a least-squares amplitude residual with unit weights. A one-bin upstream `Scaler` remains active to provide the bulk-solvent and scaler terms, and its parameters are included with the rigid-body parameters in the L-BFGS step. The target additionally applies its own detached, closed-form global work-set scale to the amplitudes it receives.

- For $d_{\min} \leq 6$ Å, the target is the standard σ_A ML target with the normal multi-bin scaler.

The switch is motivated by the noise dominance of the per-reflection σ_A target at coarse resolution: when only the lowest-resolution shells are active, the LS target with a single closed-form scale is both faster and more stable.

At coarse cutoffs an inner-cycle loop interleaves bulk-solvent mask refresh, solvent-parameter refit ($\log k_{\text{sol}}$, $\log s_h^2$, $\log n$, φ_{offset}), the rigid LBFGS step, and a reset that zeroes the Euler angles and translations while updating the reference coordinates, keeping the parameterisation in the small-angle regime (mirroring Phenix's per-macrocycle reset). All non-X-ray targets (geometry, ADP, occupancy, VDW) are dropped during the rigid step rather than zero-weighted, so the optimiser sees only the rigid-body landscape. After the final cutoff, the accumulated rotation and translation are committed back into the atomic coordinates.

12 Caching infrastructure

Many TorchRef components, the `ModelFT` structure factor forward, the parameter wrappers, and the rigid-body coordinate container, inherit a `CachedForwardMixin` that returns the previous `forward()` result when no relevant input has changed. Changes are detected by a cheap per-tensor fingerprint (each parameter, buffer, and input tracked by its `data_ptr` and version counter), and the cache is invalidated on any parameter mutation, in-place optimiser updates, freeze/unfreeze, device transfers, or on a backward pass through the cached output. Within a macrocycle's alternating blocks this lets targets skip recomputation when their inputs are untouched: F_{calc} is reused when only the scaler parameters change, and the wrappers' scattered full-array outputs are reused when only frozen selections change. The bulk-solvent mask cache uses the same fingerprinting, though it is object-specific rather than shared via the mixin.

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